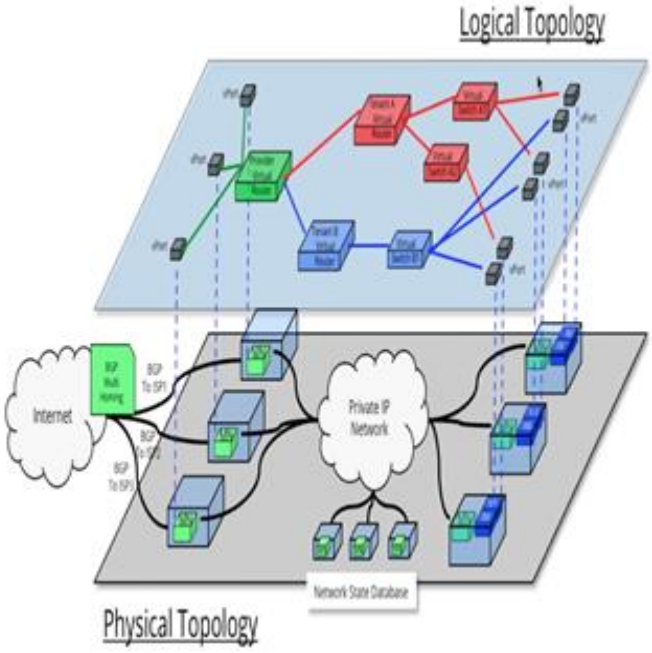


CLOUD COMPUTING COURSE

Lecture12-Network Virtualization in CC (1)

ABSTRACT

Lecture12 Introduces physical network main concepts.

Instructor: Asma'a Khtoom

In virtualization, applications run on VMs, and VMs run on physical servers. Before connecting VMs to a network, connect physical servers to the network first. To do so, the following concepts should be discussed:

1. TCP/IP Model Suite

What is a Protocol?

A protocol is a set of rules that govern how systems communicate. For networking, they govern how **data is transferred** from one system to another.

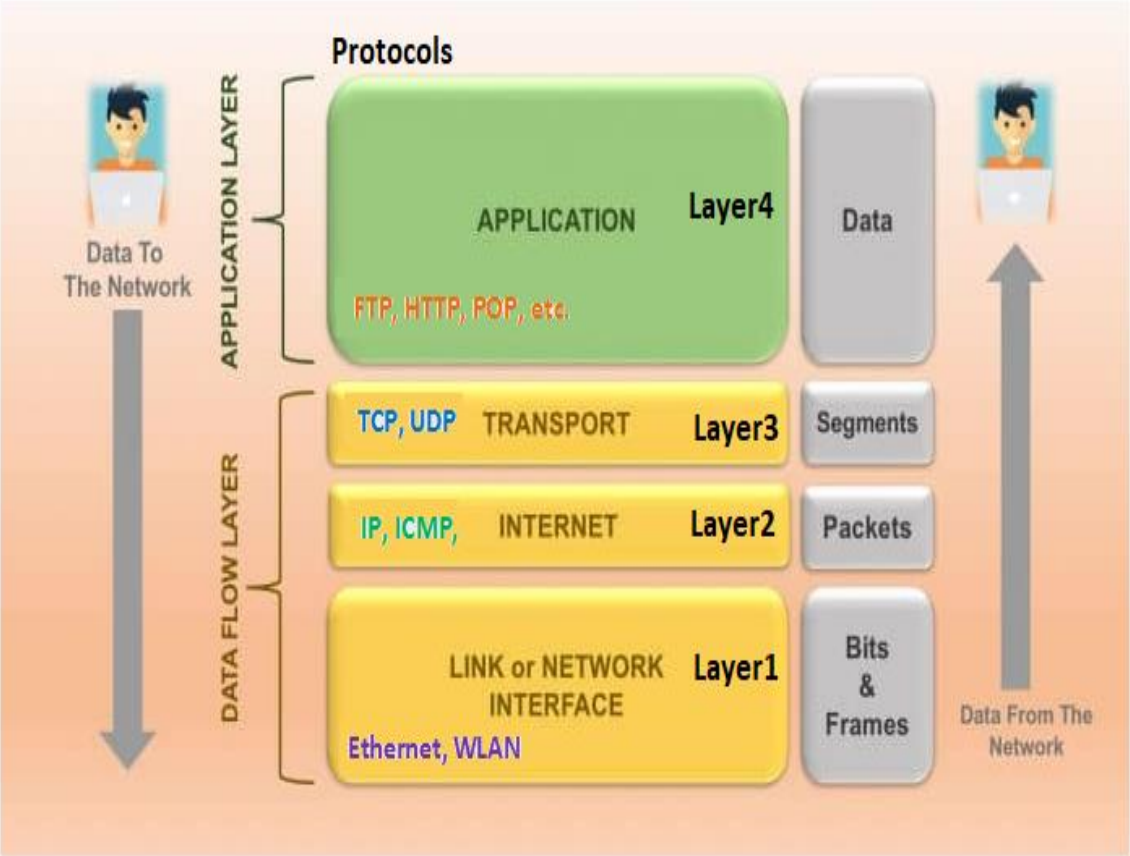
What is a Protocol Suite?

- A protocol suite is a collection of protocols that are designed to work together. The approach used in networking is to create **layered protocol stacks**.

- Each level of the stack **performs a particular function** and communicates with the levels above and below it.

- **TCP/IP** protocol suite is named after two of the most common protocols – **TCP** (transmission Control Protocol) and **IP** (internet Protocol).

- The TCP/IP protocol suite consists of many protocols that operate at one of **four layers**, as shown in the following figure.



Layer4- Application Layer:

- The topmost layer, Application Layer of TCP/IP protocol suite is concerned mainly with human interaction, the implementation of software applications and related protocols and how host programs interface with Transport layer services to use the network.
- **Application layer includes** all the higher-level protocols like **DNS** (Domain Naming System), **HTTP** (Hypertext Transfer Protocol), **Telnet**, **SSH**, **FTP** (File Transfer Protocol), **RDP** (Remote Desktop Protocol) etc.

Layer3- Transport Layer:

- It determines how much data should be sent where and at what rate. By dividing the message received from the session layer into **segments** and numbers them to make a sequence, it contains the **port number** of the sent application.

- This layer builds on the message which are received from the application layer. It helps ensure that data units are delivered error-free and in sequence.

- Transport layer helps you to control the reliability of a link through flow control, error control, and segmentation or de-segmentation.

- The **main protocols included** at Transport layer are **TCP** (Transmission Control Protocol) and **UDP** (User Datagram Protocol).

Layer2- Internet Layer:

Internet layer pack data into data packets known as IP **Packets**, which contain **source and destination address (logical address or IP address)** information that is used to forward the packets between hosts and across networks. The Internet layer is also responsible for routing of IP packets.

It send the packets from any network, and any computer still they reach the destination irrespective of the route they take.

Layer1-Link Layer /Network Access

Defines details of how data is physically sent through the network, including:

- how bits are electrically or optically signaled by hardware devices that interface directly with a network medium, such as coaxial cable, optical fiber, or twisted pair copper wire.

- The MAC addresses for both **source and destination devices** are added to the data packet to create what is called Data **Frames**.

- The protocols **included** in this Layer are Ethernet, **Token Ring**, **and Frame Relay** etc.

FOR MORE EXPLANATION, PLEASE WATCH THE RELATED VIDEOS ABOUT:

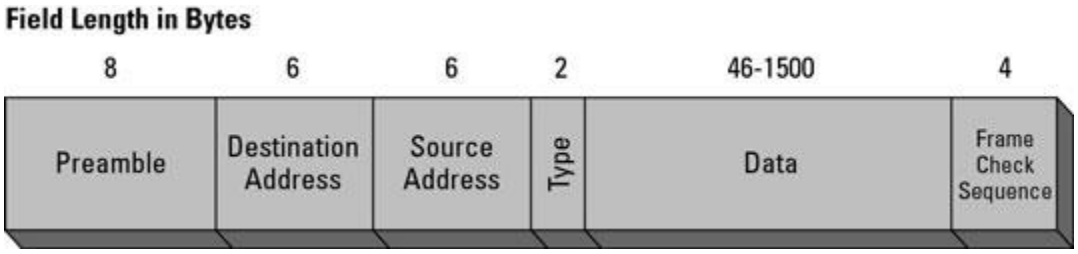
1. TCP/IP Protocol

2. TCP vs UDP

2. Ethernet Data Frames

Data travels over the physical media of the Ethernet network in small containers, or ***frames***. There are different methods of framing Ethernet data, but the two that you are likely to see are Ethernet II and IEEE 802.3. The structure of these frames is similar, and the following explains it:

As shown, the standard Ethernet frame has the following parts:



- **Preamble:** The *preamble* is a unique sequence of alternating zeros and ones followed by two ones that is eight bytes in length and sets the start of the Ethernet frame. This series of pulses is picked up by the network card in your computer and, by design, this sequence will never occur in the middle of the frame.
- **Ethernet header:** The Ethernet header is made up of the following pieces:
    - *Destination address:* Six bytes that contain the MAC address of the NIC that is the target for the network frame.
    - *Source address:* Six bytes that contain the MAC address of the NIC that sends the data on physical media.

- *Type:* Two bytes that denote the frame type. The type field identifies the higher layer protocol, which typically is IP.

- **Data:** Between 46–1500 bytes of data. If the data is fewer than 46 bytes, padding is added to bring the frame to the minimum 64 byte frame size. There must be at least 64 bytes of data between preamble sequences.
- **Frame check sequence (FCS)**: Four bytes of FCS data is stored at the end of the frame. Prior to sending the frame, the source computer generates a result from the data found in the frame and stores the result in the last four bytes of the frame.
- To generate this FCS value, the entire frame is broken into blocks. All those blocks are then added together, and FCS is a sum of all these blocks of data. The receiving computer calculates its own result from the data in the frame and compares the number it calculates to the FCS data. If the results do not match, the frame is considered to be damaged or inaccurate so the frame is discarded.

- **FOR MORE EXPLANATION, PLEASE WATCH THE RELATED VIDEOS ABOUT:**

1. 7 parts of an Ethernet Frame
2. Ethernet Frame Structure
3. Sending Ethernet Frame
4. Receiving Ethernet Frame